b) It should use a lot of storage on disk.

c) It should use a lot of network bandwidth.

d) It should use a lot of memory (several MB).

e) It should use a random salt for each password.

f) It should use a lot of CPU power to try lots of passwords.

4. What is the main reason why HTTP Basic authentication should only be used over an encrypted communication channel such as HTTPS? Pick one answer.

a) The password can be exposed in the `Referer` header.

b) HTTPS slows down attackers trying to guess passwords.

c) The password might be tampered with during transmission.

d) Google penalizes websites in search rankings if they do not use HTTPS.

e) The password can easily be decoded by anybody snooping on network traffic.

3.4 Using encryption to keep data private

Introducing authentication into your API solves spoofing threats. However, requests to the API, and responses from it, are not protected in any way, leading to tampering and